# ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 15 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 2 | **(a)** 0 | Identity. |
| 3 | **(a)** -4 | Evaluate both products. |
| 4 | **(a)** 2277 | Distributive property. |
| 5 | **(a)** 100 | Distributive property. |
| 6 | **(c)** 130 | Add the decreasing distances. |
| 7 | **(d)** 3 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 8 | **(d)** 24 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 9 | **(b)** 82 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 10 | **(a)** 0 | Identity. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Innermost brackets; Next level (1); Final operations; Value $= 15$ (1). | 1+1 |
| 12 | $= 37 \times (48 + 52)$ (1); $= 37 \times 100 = 3700$ (1). | 1+1 |
| 13 | First $= 150$; Second $= 154$ (1); First is smaller (1). | 1+1 |
| 14 | Innermost brackets; Next level (1); Final operations; Value $= 82$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Full sets $= 24 \div 6 = 4$; Discount $= 4 \times 20 = 80$ (1); Total $= 24 \times 35 + 30 \times 12 - 80 = 1120$ (1); Check with units (₹) (1). | 1+1+1 |
| 16 | First $= 180$ (1); Second $= (12 + 2)times(15 - 2) = 182$ (1); Difference $= 2$ (1). | 1+1+1 |
| 17 | Outer area $= (20 + 2 \times 2) \times (12 + 2 \times 2) = 384$ (1); Garden area $= 20 \times 12 = 240$ (1); Path area $= 144$ m$^2$ (1). | 1+1+1 |
| | **OR:** Full sets $= 30 \div 5 = 6$; Discount $= 6 \times 25 = 150$ (1); Total $= 30 \times 40 + 25 \times 15 - 150 = 1425$ (1); Check with units (₹) (1). | 1+1+1 |
| 18 | First $= 252$ (1); Second $= (14 + 3)times(18 - 3) = 255$ (1); Difference $= 3$ (1). | 1+1+1 |
| | **OR:** Outer area $= (30 + 2 \times 3) \times (18 + 2 \times 3) = 864$ (1); Garden area $= 30 \times 18 = 540$ (1); Path area $= 324$ m$^2$ (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Income $= 60 \times 3 \times 10 + 40 \times 2 \times 25$ (1); $= 3800$ (1); Expenses $= 800 + 400 = 1200$ (1); Profit $= 2600$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 21$ (1); Second $= -3$ (1); Larger: first (1); Difference 24 (2). | 1+1+1+2 |
| 20 | Income $= 50 \times 4 \times 12 + 30 \times 3 \times 20$ (1); $= 4200$ (1); Expenses $= 700 + 500 = 1200$ (1); Profit $= 3000$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 120$ (1); Second $= 40$ (1); Larger: first (1); Difference 80 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** -6 | Evaluate both products. |
| 2 | **(d)** 4455 | Distributive property. |
| 3 | **(a)** 100 | Distributive property. |
| 4 | **(a)** 94 | Add the decreasing distances. |
| 5 | **(b)** 7 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 6 | **(c)** 21 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 7 | **(d)** 32 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 8 | **(b)** 0 | Identity. |
| 9 | **(d)** -5 | Evaluate both products. |
| 10 | **(b)** 6138 | Distributive property. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $= 64 \times (27 + 73)$ (1); $= 64 \times 100 = 6400$ (1). | 1+1 |
| 12 | First $= 294$; Second $= 300$ (1); First is smaller (1). | 1+1 |
| 13 | Innermost brackets; Next level (1); Final operations; Value $= 32$ (1). | 1+1 |
| 14 | $= 25 \times (66 + 34)$ (1); $= 25 \times 100 = 2500$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Full sets $= 36 \div 6 = 6$; Discount $= 6 \times 30 = 180$ (1); Total $= 36 \times 28 + 40 \times 10 - 180 = 1228$ (1); Check with units (₹) (1). | 1+1+1 |
| 16 | First $= 130$ (1); Second $= (10 + 1)times(13 - 1) = 132$ (1); Difference $= 2$ (1). | 1+1+1 |
| 17 | Outer area $= (25 + 2 \times 2) \times (15 + 2 \times 2) = 551$ (1); Garden area $= 25 \times 15 = 375$ (1); Path area $= 176$ m$^2$ (1). | 1+1+1 |
| | **OR:** Full sets $= 20 \div 4 = 5$; Discount $= 5 \times 15 = 75$ (1); Total $= 20 \times 45 + 18 \times 20 - 75 = 1185$ (1); Check with units (₹) (1). | 1+1+1 |
| 18 | First $= 320$ (1); Second $= (16 + 4)times(20 - 4) = 320$ (1); Difference $= 0$ (1). | 1+1+1 |
| | **OR:** Outer area $= (40 + 2 \times 4) \times (24 + 2 \times 4) = 1536$ (1); Garden area $= 40 \times 24 = 960$ (1); Path area $= 576$ m$^2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Income $= 80 \times 2 \times 15 + 45 \times 2 \times 30$ (1); $= 5100$ (1); Expenses $= 900 + 600 = 1500$ (1); Profit $= 3600$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 9$ (1); Second $= 4$ (1); Larger: first (1); Difference $5$ (2). | 1+1+1+2 |
| 20 | Income $= 70 \times 3 \times 8 + 35 \times 4 \times 22$ (1); $= 4760$ (1); Expenses $= 650 + 450 = 1100$ (1); Profit $= 3660$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 4$ (1); Second $= 16$ (1); Larger: second (1); Difference $12$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Arithmetic Expressions** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 100 | Distributive property. |
| 2 | **(d)** 165 | Add the decreasing distances. |
| 3 | **(a)** 4 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 4 | **(a)** 2 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 5 | **(b)** 6 | Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction. |
| 6 | **(b)** 0 | Identity. |
| 7 | **(a)** -8 | Evaluate both products. |
| 8 | **(a)** 7722 | Distributive property. |
| 9 | **(d)** 100 | Distributive property. |
| 10 | **(d)** 140 | Add the decreasing distances. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | First $= 510$; Second $= 522$ (1); First is smaller (1). | 1+1 |
| 12 | Innermost brackets; Next level (1); Final operations; Value $= 6$ (1). | 1+1 |
| 13 | $= 18 \times (45 + 55)$ (1); $= 18 \times 100 = 1800$ (1). | 1+1 |
| 14 | First $= 216$; Second $= 221$ (1); First is smaller (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Full sets $= 48 \div 8 = 6$; Discount $= 6 \times 40 = 240$ (1); Total $= 48 \times 32 + 36 \times 14 - 240 = 1800$ (1); Check with units (₹) (1). | 1+1+1 |
| 16 | First $= 187$ (1); Second $= (11 + 2)times(17 - 2) = 195$ (1); Difference $= 8$ (1). | 1+1+1 |
| 17 | Outer area $= (36 + 2 \times 3) \times (20 + 2 \times 3) = 1092$ (1); Garden area $= 36 \times 20 = 720$ (1); Path area $= 372$ m$^2$ (1). | 1+1+1 |
| | **OR:** Full sets $= 42 \div 7 = 6$; Discount $= 6 \times 35 = 210$ (1); Total $= 42 \times 38 + 24 \times 16 - 210 = 1770$ (1); Check with units (₹) (1). | 1+1+1 |
| 18 | First $= 247$ (1); Second $= (13 + 3)times(19 - 3) = 256$ (1); Difference $= 9$ (1). | 1+1+1 |
| | **OR:** Outer area $= (45 + 2 \times 5) \times (30 + 2 \times 5) = 2200$ (1); Garden area $= 45 \times 30 = 1350$ (1); Path area $= 850$ m$^2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Income $= 90 \times 2 \times 12 + 50 \times 3 \times 18$ (1); $= 4860$ (1); Expenses $= 1000 + 700 = 1700$ (1); Profit $= 3160$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 52$ (1); Second $= 28$ (1); Larger: first (1); Difference 24 (2). | 1+1+1+2 |
| 20 | Income $= 55 \times 5 \times 6 + 25 \times 4 \times 28$ (1); $= 4450$ (1); Expenses $= 600 + 350 = 950$ (1); Profit $= 3500$ (1); Expression uses brackets correctly (1). | 1+1+1+1+1 |
| | **OR:** First $= 64$ (1); Second $= 24$ (1); Larger: first (1); Difference 40 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.